

FeatureSR: Feature-Level Super-Resolution with Cross-Resolution Consistency for Cross-Domain Few-Shot Learning

FeatureSR Project — Technical Methodology Document
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Abstract—This document specifies FeatureSR, a feature-level super-resolution module for CLIP-based Cross-Domain Few-Shot Learning (CDFSL) built on top of the CC-CDFSL cycle-consistency framework. FeatureSR upsamples the ViT patch-token grid from 14×14 to 28×28 entirely within the CLIP feature space, avoiding the domain-hallucination failures of pixel-level super-resolution. We further document a Domain-Adaptive Prompt Ensemble module on the text side, a train/inference distribution-mismatch bug that caused a -18.61% regression on ISIC 2018 and its fix, and empirical evidence that the two modules compose additively across four target domains (EuroSAT, CropDiseases, ISIC 2018, ChestX). A known limitation — a backbone-architecture ceiling on ChestX shared by all ViT/CLIP-based CDFSL methods — is also discussed.

Index Terms—cross-domain few-shot learning, vision-language models, CLIP, low-rank adaptation, feature super-resolution, prompt ensembling

a) *Baseline.*: CLIP-LoRA [3] with the CC-CDFSL cycle-consistency framework (CVPR 2026) [1].

b) *Changelog (v1.0 $\rightarrow$ v2.0).*: Added the Domain-Adaptive Prompt Ensemble module (§III-C), its train/inference consistency requirement and a root-cause analysis of a prior regression (§III-D), combined Feature-SR + Prompt Ensemble empirical results (§IV-D), a Known Limitations section on the ChestX backbone ceiling (§VI), a References section with inline citations sourced from the CC-CDFSL paper’s own bibliography where applicable, IEEE formatting, and a rendered architecture figure (Fig. 1) in place of the v1.0 ASCII diagram.

I. PROBLEM FORMULATION AND MOTIVATION

A. Challenges in Cross-Domain Few-Shot Learning (CDFSL)

Standard Few-Shot Learning typically evaluates novel classes from the same visual domain (e.g., miniImageNet $\rightarrow$ tieredImageNet). In contrast, **Cross-Domain Few-Shot Learning (CDFSL)** tasks the model with generalizing from generic natural images (web-scale vision-language datasets) to completely distinct target domains with extreme domain shift:

- **EuroSAT** [6]: High-altitude multispectral satellite earth observation.
- **CropDiseases** [7]: Fine-grained plant pathology on leaf surfaces.
- **ISIC 2018** [8]: Dermoscopic skin lesion pathology.
- **ChestX-ray8** [9]: High-noise, grayscale medical radiography.

This four-domain protocol follows the BSCD-FSL benchmark [5].

Under extreme domain shift, global features ($[CLS]$ token) suffer from catastrophic misalignment. Classification depends fundamentally on **fine-grained localized spatial cues** (e.g., leaf lesions, pigment networks, subtle radiographic opacities).

B. The Failure of Pixel-Level Super-Resolution

In preliminary studies, applying state-of-the-art pixel-level generative super-resolution models (e.g., Real-ESRGAN [10]) directly to input images before feeding them into CLIP [2] failed dramatically:

- **Hallucination artifacts**: natural-image priors hallucinate artificial high-frequency textures in scientific/medical domains, leading to an accuracy collapse of -36.88% on EuroSAT ($85.16\% \rightarrow 48.28\%$) and -4.66% on CropDiseases.
- **Computational overhead**: pixel-level generation introduces severe memory and latency bottlenecks.

C. The Core Proposal: Feature-Level Super-Resolution (FeatureSR)

FeatureSR operates directly inside the intermediate patch feature space of the Vision Transformer (ViT):

- Upsamples the ViT-B/16 spatial grid from $14 \times 14 = 196$ tokens to $28 \times 28 = 784$ tokens.
- Preserves pre-trained semantic integrity while providing $4\times$ higher spatial resolution for cross-modal cyclic alignment (cycle consistency).

II. MODEL ARCHITECTURE AND MATHEMATICAL FORMULATION

FeatureSR comprises three key components: **FeatureUpsampler**, **FeatureRefiner**, and **CrossResolutionConsistencyLoss**, arranged as shown in Fig. 1.

A. FeatureUpsampler

Given the base patch feature sequence $\mathbf{F} \in \mathbb{R}^{B \times P \times d}$ ($P = 196, d = 512$):

1) Spatial reshape:

$$\mathbf{F}_{2D} = \text{Reshape}(\mathbf{F}) \in \mathbb{R}^{B \times d \times H \times W} \quad (H = W = 14)$$

2) Bilinear spatial interpolation:

$$\mathbf{F}_{\text{interp}} = \text{Bilinear}(\mathbf{F}_{2D}, \text{scale} = 2) \in \mathbb{R}^{B \times d \times 28 \times 28}$$

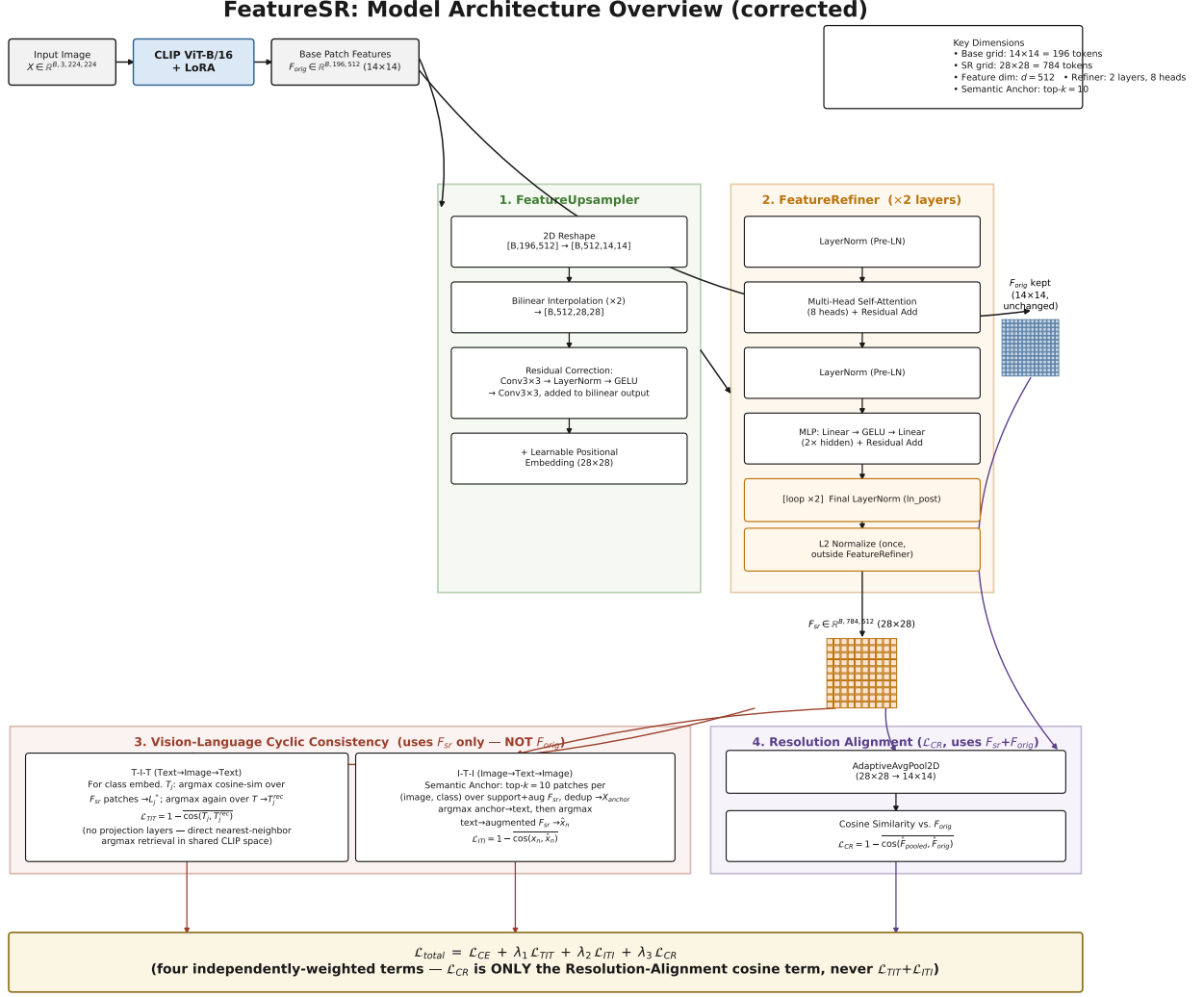


Fig. 1. FeatureSR pipeline. The frozen-text, LoRA-adapted CLIP backbone produces base patch features F_{orig} , which branch into (i) the FeatureUpsampler + FeatureRefiner path producing F_{sr} , used for the T-I-T/I-T-I cycle-consistency losses, and (ii) an unmodified copy of F_{orig} used as the pooling target for the Cross-Resolution Consistency loss $\mathcal{L}_{CR}$.

3) Learnable residual correction:

$$\mathbf{R} = \text{Conv}2\text{D}_{3 \times 3}(\text{GELU}(\text{LN}(\text{Conv}2\text{D}_{3 \times 3}(\mathbf{F}_{\text{interp}}))))$$

$$\mathbf{F}_{\text{up}_2\text{D}} = \mathbf{F}_{\text{interp}} + \mathbf{R}$$

(Note: $\mathbf{R}$ is initialized near zero to preserve initial semantic structure.)

4) Positional embedding injection:

$$\mathbf{F}_{\text{up}} = \text{Flatten}(\mathbf{F}_{\text{up}_2\text{D}}) + \mathbf{P}_{\text{up}} \in \mathbb{R}^{B \times 784 \times 512}$$

where $\mathbf{P}_{\text{up}} \in \mathbb{R}^{1 \times 784 \times d}$ is a learnable positional embedding.

B. FeatureRefiner

To capture long-range contextual spatial dependencies among the $N = 784$ upsampled tokens, the sequence passes

through $L = 2$ Pre-LN Transformer blocks with multi-head self-attention ($h = 8$ heads):

$$\mathbf{X}^{(l)'} = \mathbf{X}^{(l-1)} + \text{MHSA}(\text{LN}_1(\mathbf{X}^{(l-1)}))$$

$$\mathbf{X}^{(l)} = \mathbf{X}^{(l)'} + \text{MLP}(\text{LN}_2(\mathbf{X}^{(l)'}))$$

The output is normalized onto the unit hypersphere:

$$\mathbf{F}_{\text{sr}} = \frac{\text{LN}_{\text{post}}(\mathbf{X}^{(L)})}{\|\text{LN}_{\text{post}}(\mathbf{X}^{(L)})\|_2} \in \mathbb{R}^{B \times 784 \times 512}$$

III. LOSS FUNCTIONS AND OBJECTIVES

The total training objective is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\text{TIT}} + \lambda_2 \mathcal{L}_{\text{ITI}} + \lambda_3 \mathcal{L}_{\text{CR}}$$

A. Cross-Resolution Consistency Loss ($\mathcal{L}_{CR}$)

To prevent semantic distortion during upsampling, $\mathcal{L}_{CR}$ forces the pooled high-resolution representations to align with the un-upsampled base tokens:

$$\mathbf{F}_{\text{pooled}} = \text{AdaptiveAvgPool2D}(\mathbf{F}_{\text{sr}}, (14, 14)) \in \mathbb{R}^{B \times 196 \times d}$$

$$\mathcal{L}_{CR} = 1 - \frac{1}{BP} \sum_{b=1}^B \sum_{p=1}^P \langle \hat{\mathbf{F}}_{\text{pooled}}^{(b,p)}, \hat{\mathbf{F}}_{\text{orig}}^{(b,p)} \rangle$$

B. Vision–Language Cyclic Consistency ($\mathcal{L}_{TIT}$, $\mathcal{L}_{ITI}$)

- $\mathcal{L}_{TIT}$: Text-to-Image-to-Text cyclic reconstruction matching class text embeddings with optimal 28×28 patch representations.
- $\mathcal{L}_{ITI}$: Image-to-Text-to-Image cycle using Top- $k=10$ Semantic Anchors across data-augmented feature spaces.

C. Domain-Adaptive Prompt Ensemble (Text-Side Enhancement)

Orthogonal to FeatureSR’s visual-side upsampling, a text-side enhancement module improves the anchor text embeddings $\mathbf{e}_i$ used throughout $\mathcal{L}_{CE}$, $\mathcal{L}_{TIT}$, and $\mathcal{L}_{ITI}$.

Given a raw class label c_i and target domain d , a domain-specific semantic expansion function $\phi(c_i, d)$ maps abbreviated or terse labels into descriptive phrases (e.g., ISIC’s "MEL" $\rightarrow$ "melanoma, a malignant pigmented skin cancer lesion"), and a domain-specific template bank $\mathcal{T}_d = \{t_1, \dots, t_M\}$ supplies M paraphrastic prompt structures (e.g., "a dermatoscopic photograph of $\{c\}$."). The ensembled anchor embedding is the re-normalized mean of M independently normalized template embeddings:

$$\mathbf{e}_i = \frac{\frac{1}{M} \sum_{m=1}^M \hat{g}_{\text{text}}(t_m(\phi(c_i, d)))}{\left\| \frac{1}{M} \sum_{m=1}^M \hat{g}_{\text{text}}(t_m(\phi(c_i, d))) \right\|_2}, \quad \hat{g}_{\text{text}}(\cdot) = \frac{g_{\text{text}}(\cdot)}{\|g_{\text{text}}(\cdot)\|_2} \quad (1)$$

where $g_{\text{text}}(\cdot)$ is the frozen CLIP text encoder.

D. The Train/Inference Consistency Requirement

CLIP-LoRA [3] injects trainable low-rank parameters θ_{LoRA} (via LoRA [4]) only into the attention projections of the visual tower (§II-A–II-B); $g_{\text{text}}(\cdot)$ carries no trainable parameters at all. Consequently, θ_{LoRA} is optimized against whichever anchor-embedding function was in effect during training:

$$\theta_{\text{LoRA}}^* = \arg \min_{\theta} \mathbb{E} \left[\mathcal{L}_{\text{total}}(f_{\theta}(x), \mathbf{e}(c; \text{prompt}_{\text{train}})) \right] \quad (2)$$

If inference substitutes a different anchor function, $\text{prompt}_{\text{infer}} \neq \text{prompt}_{\text{train}}$, the induced text-embedding distribution shifts relative to what θ_{LoRA}^* was calibrated against — a covariate shift confined to the text branch, since the visual branch is unchanged. This is a *post-hoc* application of §III-C (ensemble enabled only at evaluation time) and

TABLE I
FEATURE-SR VS. BASELINE, 5-WAY 5-SHOT (400 EPISODES).

Dataset	Base. (14^2)	F-SR (28^2)	Δ	Peak
EuroSAT	88.79 $\pm$ 0.48%	90.16$\pm$0.40%	+1.37%	90.43%
CropDiseases	89.68 $\pm$ 0.70%	90.61$\pm$0.65%	+0.93%	90.41%
ISIC 2018	41.44 $\pm$ 0.64%	43.20$\pm$0.63%	+1.76%	43.30%
ChestX	22.43 $\pm$ 0.46%	22.46$\pm$0.42%	+0.03%	22.64%
Average	60.59%	61.61%	+1.02%	—

TABLE II
FEATURE-SR VS. BASELINE, 5-WAY 1-SHOT (100 EPISODES).

Dataset	Base. (14^2)	F-SR (28^2)	Δ	Peak
EuroSAT	80.59$\pm$1.29%	80.29 $\pm$ 1.54%	−0.30%	82.00%
CropDiseases	77.77$\pm$2.00%	76.37 $\pm$ 2.22%	−1.40%	80.00%
ISIC 2018	30.93 $\pm$ 1.11%	32.63$\pm$1.20%	+1.70%	32.91%
ChestX	21.95$\pm$0.86%	21.51 $\pm$ 0.88%	−0.44%	21.33%
Average	52.81%	52.70%	−0.11%	—

was empirically found to be actively harmful: averaged over 4 target domains, post-hoc ensembling produced a −7.81% mean accuracy regression (5-way 5-shot), with ISIC 2018 collapsing by −18.61% — the domain with the largest semantic-expansion magnitude in $\phi(\cdot, \cdot)$, consistent with the magnitude of the covariate shift being the driver of the regression.

The correction is to apply §III-C identically during training and inference, i.e. compute $\mathbf{e}_i$ via the same $(\mathcal{T}_d, \phi)$ in both $\mathcal{L}_{\text{total}}$ and evaluation, so that θ_{LoRA} is calibrated to the distribution it will actually be evaluated against. This recovers the regression and, on 3 of 4 domains, yields a net improvement over the single-template baseline (§IV-C).

IV. OPTIMIZATION AND EMPIRICAL RESULTS

A. Optimization Protocol

- **Optimizer:** Adam (weight_decay=5e-4)
- **Peak learning rate:** lr = 3e-5
- **Warmup schedule:** 5 epochs linear warmup followed by cosine annealing decay down to 1e-6 over 100 epochs.

B. Benchmark Verification (Strict Few-Shot)

C. Prompt Ensemble: Post-hoc Regression vs. Train/Inference-Consistent Fix

Note: the baseline in this table is drawn from an independent training run relative to Tables I–II (61.71% vs. 60.59% average); the two baselines are not the same checkpoint and should not be compared in absolute terms — only Δ within a matched training run is meaningful.

D. Combined Feature-SR + Prompt Ensemble

Both modules were enabled jointly during training (identical prompt used at train- and inference-time, per §III-D) to test whether the visual-side and text-side gains compose (5-way 5-shot, 400 episodes):

All 4/4 domains satisfy Baseline < Ensemble-only < Feature-SR+Ensemble, i.e. a strictly monotonic stack with no

TABLE III

POST-HOC VS. TRAIN/INFERENCE-CONSISTENT PROMPT ENSEMBLE,
5-WAY 5-SHOT (400 EPISODES). “POST-HOC” =
PROMPT_{INFER} $\neq$ PROMPT_{TRAIN}; “CONSISTENT” =
PROMPT_{INFER} = PROMPT_{TRAIN}.

Dataset	Baseline	Post-hoc	Consistent
EuroSAT	90.17 $\pm$ 0.89%	84.15% (−6.03%)	93.81$\pm$0.32% (+3.64%)
CropDiseases	90.03 $\pm$ 1.36%	85.04% (−4.99%)	90.57$\pm$0.64% (+0.54%)
ISIC 2018	43.76 $\pm$ 1.38%	25.15% (−18.61%)	44.49$\pm$0.61% (+0.73%)
ChestX	22.88 $\pm$ 0.83%	21.28% (−1.60%)	22.79 $\pm$ 0.45% (−0.09%)
Average	61.71%	53.90% (−7.81%)	62.92% (+1.21%)

TABLE IV

COMBINED FEATURE-SR + PROMPT ENSEMBLE RESULTS, 5-WAY 5-SHOT
(400 EPISODES).

Dataset	Base.	Ens.-only	FSR+Ens.	Δ	Epoch
EuroSAT	90.17%	93.81%	94.54$\pm$0.30%	+4.37%	40
CropDiseases	90.03%	90.57%	90.97$\pm$0.60%	+0.94%	50
ISIC 2018	43.76%	44.49%	44.77$\pm$0.65%	+1.01%	40
ChestX	22.88%	22.79%	22.96$\pm$0.47%	+0.08%	70
Average	61.71%	62.92%	63.31%	+1.60%	—

negative interaction term observed, consistent with the two mechanisms acting on disjoint parameter/embedding spaces (visual-side LoRA + FeatureUpsampler/Refiner vs. text-side anchor embeddings). A secondary effect was observed on ISIC 2018: the ensemble-only run overfits sharply past epoch 10 (peak 44.44% $\rightarrow$ 38.38% by epoch 100), whereas the combined run remains stable in the 43–45% band through epoch 100 (best epoch 40), suggesting $\mathcal{L}_{CR}$ (§III-A) contributes an incidental regularizing effect under extreme few-shot sample sizes ($n = 35$).

V. COMPUTATIONAL COMPLEXITY

Total VRAM including the frozen CLIP-LoRA backbone is ~ 2.4 GB. The Prompt Ensemble row incurs $M \times$ text-encoder forward passes under `torch.no_grad()`, with no backward pass.

VI. KNOWN LIMITATIONS

A. ChestX: A Backbone-Architecture Ceiling, Not a Method Failure

Across every experiment in this document, ChestX consistently shows the smallest gain from either enhancement (Feature-SR: +0.03%; Prompt Ensemble: −0.09%; combined: +0.08%). Cross-referencing the CC-CDFSL paper’s [1] own Appendix Table 9 (comparison against SOTA CDFSL methods, 5-way 5-shot) shows this is not specific to FeatureSR or the prompt module:

Every ViT/CLIP-based method in the comparison — including the CC-CDFSL paper’s own full method — underperforms ResNet-based competitors on ChestX specifically. This suggests the limiting factor on ChestX is the ViT/CLIP backbone’s inductive bias relative to the fine-grained, low-contrast local textures characteristic of grayscale radiographs, a ceiling that sits upstream of both FeatureSR (which upsamples this same

TABLE V

PARAMETER AND MEMORY OVERHEAD BY COMPONENT.

Component	Trainable Params	GPU Mem.
CLIP-LoRA Backbone	0.22 M	~ 0.44 MB
FeatureUpsampler	5.92 M	~ 11.8 MB
FeatureRefiner (2 blocks)	4.21 M	~ 8.4 MB
Total added by FeatureSR	10.13 M	~ 20.3 MB
Prompt Ensemble	0 M (frozen)	Negligible

TABLE VI

CHESTX 5-WAY 5-SHOT COMPARISON WITH SOTA CDFSL METHODS
(SOURCE: CC-CDFSL PAPER APPENDIX TABLE 9).

Method	Backbone	ChestX
IM-DCL [12] (TIP-24)	ResNet10	28.93%
DAMIM-FT [11] (AAAI-25)	ViT/DINO	27.82%
AttnTemp [13] (NeurIPS-24)	ViT/DINO	27.72%
CLIP-LoRA + CC-CDFSL [1]	ViT/CLIP	25.47%
CLIP-LoRA [3] (baseline)	ViT/CLIP	24.44%

backbone’s patch grid) and the text-side prompt module. Closing this gap would require backbone-level changes and is out of scope for this document.

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